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Project Overview and Goals:

This project aimed to create a predictive linear regression model that can take a country's predicted energy consumption and renewable energy percentage to predict carbon footprint for a future year. The country chosen for running queries is the United States but the prediction SQL command can be modified to be any Entity country along with carrying values for fossil_fuel_usage, renewable_usage, renewable_percentage, and annual_temp_anomaly.

Datasets:

https://www.kaggle.com/datasets/anshtanwar/global-data-on-sustainable-energy https://data.giss.nasa.gov/gistemp/tabledata_v4/GLB.Ts+dSST.csv

Description: Structured CSV with 3650 rows and 21 columns spanning from 2000 – 2020. Some countries have no reported values in some columns, mostly surrounding renewable energy. Global-mean monthly, seasonal, and annual means from 1880 – present holding temperature anomalies.

Pipeline and Services:

Google Cloud Storage, BigQuery and BigQuery ML, and Looker Studio were used for the project. The files were loaded into a cloud storage bucket within the cloud project and transformed using BigQuery. A BigQuery ML linear regression model was created and connected to Looker Studio as a data source for charting.

Data Processing:

The NASA dataset had to have 2 rows skipped but the schema was kept as generated in order to successfully join. Both files were CSV so it was not very difficult to process them.

Results:
The “full” table was created from joining both datasets after they were cleaned and ready to join. R squared value was very high when evaluating the model.



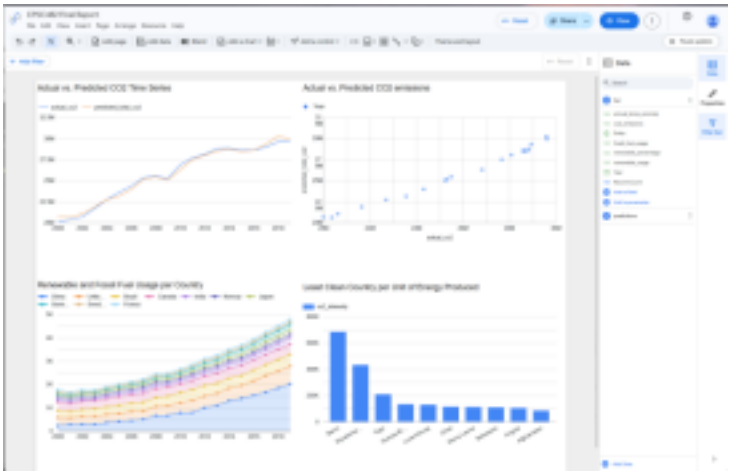
jou-cpsc482-finalproject / Datasets / project_sets / Models / country_prediction_model

☆ country_prediction_model Refresh

Details | Training metrics | Evaluation | Inference | Warm-start | Registry

Mean absolute error	11,378.7084
Mean squared error	1,566,573,739.2543
Mean squared log error	0.6231
Median absolute error	2,388.0963
R squared	0.9981

Visualizations:
https://datastudio.google.com/reporting/a7ed0f13-70c4-42ea-accc2522085b1faf/page/lyixF/edit



Challenges:

The NASA dataset contained its title in the table before the actual headers, so I had to skip the first 2 rows in a reuploaded table to fix it. The sustainable energy dataset had some asterisks meaning null values. The character mapping kept misinterpreting parentheses in column names for both tables.